Daily Problem Solutions Booklet

Comprehensive Step-by-Step Analytical Solutions

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Contents

Integrals

1.1 Medium Integral

Substitution and Integration by Parts

Evaluate the definite integral:

$$\int_0^{\frac{e^2-1}{2e}} \frac{\ln(x+\sqrt{x^2+1}) \cdot (x+\sqrt{x^2+1})^3}{\sqrt{x^2+1}}\, dx$$

Solution: Let the target integral be denoted by I. The presence of $\ln(x+\sqrt{x^2+1})$ strongly suggests substituting the inverse hyperbolic sine function, $u = \operatorname{arsinh}(x) = \ln(x+\sqrt{x^2+1})$. Differentiating u with respect to x yields:

$$du = \frac{1}{\sqrt{x^2+1}}\, dx$$

Additionally, by the definition of u, we can express the exponential as:

$$e^u = x + \sqrt{x^2+1}$$

Now we determine the new limits of integration. For the lower limit, when $x = 0$:

$$u = \ln(0+\sqrt{0+1}) = \ln(1) = 0$$

For the upper limit, when $x = \frac{e^2-1}{2e} = \frac{e-e^{-1}}{2} = \sinh(1)$:

$$u = \operatorname{arsinh}(\sinh(1)) = 1$$

Applying these transformations, the integral simplifies considerably:

$$I = \int_0^1 u(e^u)^3\, du = \int_0^1 ue^{3u}\, du$$

This is a standard integral that can be evaluated using integration by parts. Let $w = u$ and $dv = e^{3u}\, du$. Then, $dw = du$ and $v = \frac{1}{3}e^{3u}$. Applying the integration by parts formula $\int w\, dv = wv - \int v\, dw$:

$$I = \left[\frac{u}{3}e^{3u}\right]_0^1 - \int_0^1 \frac{1}{3}e^{3u}\, du$$

$$I = \frac{1}{3}e^3 - \left[\frac{1}{9}e^{3u}\right]_0^1 = \frac{1}{3}e^3 - \left(\frac{1}{9}e^3 - \frac{1}{9}\right)$$

Combining the terms yields the exact value:

$$I = \frac{2}{9}e^3 + \frac{1}{9} = \frac{2e^3+1}{9}$$

1.2 Hard Integral

Trigonometric Fractional Integral

Evaluate the definite integral:

$$\int_0^{\frac{\pi}{2}} \frac{\sqrt{\sin^3(x)\cos(x)}}{1+\sin(x)\cos(x)}\,dx$$

Solution: Let the integral be denoted by I. We can rewrite the integrand by recognizing symmetry. Let's employ the property $\int_0^a f(x)\,dx = \int_0^a f(a-x)\,dx$. Substituting $x \to \frac{\pi}{2} - x$ gives:

$$I = \int_0^{\frac{\pi}{2}} \frac{\sqrt{\cos^3(x)\sin(x)}}{1+\cos(x)\sin(x)}\,dx$$

Adding the two representations of I together:

$$2I = \int_0^{\frac{\pi}{2}} \frac{\sqrt{\sin(x)\cos(x)}(\sin(x)+\cos(x))}{1+\sin(x)\cos(x)}\,dx$$

We apply the substitution $t = \sin(x) - \cos(x)$, which gives $dt = (\cos(x)+\sin(x))\,dx$. Squaring the substitution gives $t^2 = \sin^2(x) + \cos^2(x) - 2\sin(x)\cos(x) = 1 - 2\sin(x)\cos(x)$, from which we deduce $\sin(x)\cos(x) = \frac{1-t^2}{2}$. The limits change from $x \in [0, \frac{\pi}{2}]$ to $t \in [-1, 1]$. The integral becomes:

$$2I = \int_{-1}^{1} \frac{\sqrt{\frac{1-t^2}{2}}}{1+\frac{1-t^2}{2}}\,dt = \sqrt{2}\int_{-1}^{1} \frac{\sqrt{1-t^2}}{3-t^2}\,dt$$

Since the integrand is even, we can write:

$$2I = 2\sqrt{2}\int_0^1 \frac{\sqrt{1-t^2}}{3-t^2}\,dt$$

Let $t = \sin(\theta)$, $dt = \cos(\theta)\,d\theta$. The limits become 0 to $\frac{\pi}{2}$:

$$2I = 2\sqrt{2}\int_0^{\frac{\pi}{2}} \frac{\cos^2(\theta)}{3-\sin^2(\theta)}\,d\theta$$

Divide numerator and denominator by $\cos^2(\theta)$, recognizing $d\theta = \frac{du}{1+u^2}$ where $u = \tan(\theta)$:

$$\int_0^{\frac{\pi}{2}} \frac{1}{3\sec^2(\theta)-\tan^2(\theta)}\,d\theta = \int_0^{\infty} \frac{1}{(3(1+u^2)-u^2)(1+u^2)}\,du = \int_0^{\infty} \frac{1}{(3+2u^2)(1+u^2)}\,du$$

Using partial fractions:

$$\frac{1}{(3+2u^2)(1+u^2)} = \frac{1}{1+u^2} - \frac{2}{3+2u^2}$$

Integrating term by term from 0 to ∞:

$$\int_0^{\infty} \frac{1}{1+u^2}\,du = [\arctan(u)]_0^{\infty} = \frac{\pi}{2}$$

$$\int_0^\infty \frac{2}{3+2u^2}\,du = \frac{2}{3}\int_0^\infty \frac{1}{1+\left(\sqrt{\frac{2}{3}}u\right)^2}\,du = \frac{2}{3}\sqrt{\frac{3}{2}}\left[\arctan\left(\sqrt{\frac{2}{3}}u\right)\right]_0^\infty = \sqrt{\frac{2}{3}}\frac{\pi}{2}$$

Putting it all together for $2I$:

$$2I = 2\sqrt{2}\left(\frac{\pi}{2} - \sqrt{\frac{2}{3}}\frac{\pi}{2}\right) = \pi\sqrt{2} - \pi\frac{2}{\sqrt{3}}$$

Dividing by 2 to isolate I:

$$I = \frac{\pi\sqrt{2}}{2} - \frac{\pi\sqrt{3}}{3}$$

Differentials

2.1 Medium Differential

Second-Order Linear Non-Homogeneous Differential Equation

Let y satisfy the following differential equation:

$$y'' + 9y = \cosh(x)$$

Given that $y(0) = 0$ and $y'(0) = 2$, find the exact value of $y(e)$.

Solution: We first solve the corresponding homogeneous equation $y_h'' + 9y_h = 0$. The characteristic equation is $r^2 + 9 = 0$, giving purely imaginary roots $r = \pm 3i$. The homogeneous solution is:

$$y_h(x) = C_1 \cos(3x) + C_2 \sin(3x)$$

Next, we find a particular solution $y_p(x)$ to the non-homogeneous equation. The right-hand side is $\cosh(x)$. We guess a solution of the form:

$$y_p(x) = A\cosh(x) + B\sinh(x)$$

Differentiating twice, $y_p''(x) = A\cosh(x) + B\sinh(x)$. Substituting this into the differential equation:

$$(A\cosh(x) + B\sinh(x)) + 9(A\cosh(x) + B\sinh(x)) = \cosh(x)$$

$$10A\cosh(x) + 10B\sinh(x) = \cosh(x)$$

Matching coefficients, we find $10A = 1 \implies A = \frac{1}{10}$ and $10B = 0 \implies B = 0$. The particular solution is $y_p(x) = \frac{1}{10}\cosh(x)$. The general solution is therefore:

$$y(x) = C_1 \cos(3x) + C_2 \sin(3x) + \frac{1}{10}\cosh(x)$$

Now, we apply the initial conditions to determine the constants C_1 and C_2.

$$y(0) = C_1 \cos(0) + C_2 \sin(0) + \frac{1}{10}\cosh(0) = C_1 + \frac{1}{10} = 0 \implies C_1 = -\frac{1}{10}$$

Differentiating the general solution to use the second initial condition:

$$y'(x) = -3C_1 \sin(3x) + 3C_2 \cos(3x) + \frac{1}{10}\sinh(x)$$

$$y'(0) = 3C_2 \cos(0) + \frac{1}{10}\sinh(0) = 3C_2 = 2 \implies C_2 = \frac{2}{3}$$

The particular solution is:

$$y(x) = -\frac{1}{10}\cos(3x) + \frac{2}{3}\sin(3x) + \frac{1}{10}\cosh(x)$$

Evaluating this solution at $x = e$:

$$y(e) = -\frac{1}{10}\cos(3e) + \frac{2}{3}\sin(3e) + \frac{1}{10}\cosh(e)$$

2.2 Hard Differential

Partial Differential Equation via Laplace Transform

A homogeneous nano-rod of length $L = 6$ m is modeled by:

$$\frac{\partial^2 \theta}{\partial x^2} + \beta^2 x = \frac{1}{a^2}\frac{\partial \theta}{\partial t}$$

Initial/boundary conditions: $\theta(x, 0) = 0$, $\theta(0, t) = 0$, $\frac{\partial \theta}{\partial x}(L, t) = 0$. Using the Laplace transform method, determine the steady-state temperature distribution $\theta(x, \infty)$. Evaluate $\theta(3, \infty)$ at $\beta = 2$.

Solution: Let $\Theta(x, s) = \mathcal{L}\{\theta(x, t)\}$. Transforming the PDE with respect to t:

$$\frac{\partial^2 \Theta}{\partial x^2} + \frac{\beta^2 x}{s} = \frac{1}{a^2}(s\Theta(x, s) - \theta(x, 0)) = \frac{s}{a^2}\Theta(x, s)$$

This forms a non-homogeneous ordinary differential equation in x:

$$\frac{\partial^2 \Theta}{\partial x^2} - \frac{s}{a^2}\Theta = -\frac{\beta^2 x}{s}$$

The homogeneous solution is $\Theta_h(x, s) = A(s)\cosh\left(\frac{\sqrt{s}}{a}x\right) + B(s)\sinh\left(\frac{\sqrt{s}}{a}x\right)$. A particular solution is of the form $\Theta_p = cx$. Substituting this gives $-\frac{s}{a^2}(cx) = -\frac{\beta^2 x}{s} \implies c = \frac{a^2\beta^2}{s^2}$. The general solution in the Laplace domain is:

$$\Theta(x, s) = A(s)\cosh\left(\frac{\sqrt{s}}{a}x\right) + B(s)\sinh\left(\frac{\sqrt{s}}{a}x\right) + \frac{a^2\beta^2}{s^2}x$$

Applying boundary conditions: $\theta(0, t) = 0 \implies \Theta(0, s) = 0 \implies A(s) = 0$. $\frac{\partial \Theta}{\partial x} = B(s)\frac{\sqrt{s}}{a}\cosh\left(\frac{\sqrt{s}}{a}x\right) + \frac{a^2\beta^2}{s^2}$. At $x = L$: $\frac{\partial \Theta}{\partial x}(L, s) = 0 \implies B(s) = -\frac{a^3\beta^2}{s^{5/2}\cosh\left(\frac{\sqrt{s}}{a}L\right)}$.

$$\Theta(x, s) = \frac{a^2\beta^2}{s^2}x - \frac{a^3\beta^2\sinh\left(\frac{\sqrt{s}}{a}x\right)}{s^{5/2}\cosh\left(\frac{\sqrt{s}}{a}L\right)}$$

To find the steady-state $\theta_s(x) = \theta(x, \infty)$, we use the Final Value Theorem: $\theta_s(x) = \lim_{s\to 0} s\Theta(x, s)$.

$$s\Theta(x, s) = \frac{a^2\beta^2 x}{s} - \frac{a^3\beta^2\sinh\left(\frac{\sqrt{s}}{a}x\right)}{s^{3/2}\cosh\left(\frac{\sqrt{s}}{a}L\right)}$$

We expand the hyperbolic functions around $s = 0$: $\sinh(z) \approx z + \frac{z^3}{6}$ and $\cosh(z) \approx 1 + \frac{z^2}{2}$. Numerator of the second term: $a^3\beta^2\left(\frac{\sqrt{s}}{a}x + \frac{s^{3/2}}{6a^3}x^3\right) = \frac{a^2\beta^2 x}{s^{-1/2}} + \frac{\beta^2 x^3}{6}s^{3/2}$. Dividing by $s^{3/2}\cosh\left(\frac{\sqrt{s}L}{a}\right) \approx s^{3/2}\left(1 + \frac{sL^2}{2a^2}\right)$, the term becomes:

$$\frac{\frac{a^2\beta^2 x}{s} + \frac{\beta^2 x^3}{6}}{1 + \frac{sL^2}{2a^2}} \approx \left(\frac{a^2\beta^2 x}{s} + \frac{\beta^2 x^3}{6}\right)\left(1 - \frac{sL^2}{2a^2}\right) \approx \frac{a^2\beta^2 x}{s} - \frac{\beta^2 x L^2}{2} + \frac{\beta^2 x^3}{6}$$

Subtracting this from the first term $\frac{a^2\beta^2 x}{s}$:

$$\theta_s(x) = \lim_{s\to 0} s\Theta(x,s) = \frac{\beta^2 x L^2}{2} - \frac{\beta^2 x^3}{6} = \frac{\beta^2}{6}(3L^2 x - x^3)$$

Evaluating at $x = 3$, $L = 6$, $\beta = 2$:

$$\theta_s(3) = \frac{2^2}{6}(3(6^2)(3) - 3^3) = \frac{2}{3}(324 - 27) = \frac{2}{3}(297) = 198$$

Derivatives

3.1 Medium Derivative

Cubic Polynomial Evaluation

Given a cubic polynomial $f(x)$ where:

$$f(2) = 3, \quad f'(2) = 4, \quad f''(2) = 14, \quad f'''(2) = 12$$

Find $f(2) + f(0) + f(2) + f(6)$.

Solution: Since $f(x)$ is a cubic polynomial, it is completely determined by its Taylor series expansion around $x = 2$, which terminates after the third derivative:

$$f(x) = f(2) + f'(2)(x-2) + \frac{f''(2)}{2!}(x-2)^2 + \frac{f'''(2)}{3!}(x-2)^3$$

Substituting the given derivative values:

$$f(x) = 3 + 4(x-2) + \frac{14}{2}(x-2)^2 + \frac{12}{6}(x-2)^3$$

$$f(x) = 3 + 4(x-2) + 7(x-2)^2 + 2(x-2)^3$$

We now evaluate the polynomial at the requested discrete values. We already know $f(2) = 3$. For $x = 0$:

$$f(0) = 3 + 4(-2) + 7(-2)^2 + 2(-2)^3 = 3 - 8 + 7(4) + 2(-8) = -5 + 28 - 16 = 7$$

For $x = 6$:

$$f(6) = 3 + 4(4) + 7(4)^2 + 2(4)^3 = 3 + 16 + 7(16) + 2(64) = 19 + 112 + 128 = 259$$

Summing these values together as requested in the problem statement:

$$f(2) + f(0) + f(2) + f(6) = 3 + 7 + 3 + 259 = 272$$

3.2 Hard Derivative

Infinite Product Derivative

Let $f(x)$ be a function defined for all $x \in \mathbb{R}$ by the infinite product:

$$f(x) = \prod_{n=1}^{\infty} \left(1 - \frac{x^2}{n^2}\right)$$

Compute the exact value of $f''(1)$.

Solution: We recognize the infinite product as Euler's famous product formula for the sine function. The un-normalized sinc function has the product representation:

$$\frac{\sin(\pi x)}{\pi x} = \prod_{n=1}^{\infty}\left(1 - \frac{x^2}{n^2}\right)$$

Therefore, our function is $f(x) = \frac{\sin(\pi x)}{\pi x}$. To find $f''(1)$, an elegant approach is to compute the Taylor series expansion of $f(x)$ centered at $x = 1$. Let $x = 1 + u$, where $u \to 0$.

$$f(1+u) = \frac{\sin(\pi(1+u))}{\pi(1+u)} = \frac{\sin(\pi + \pi u)}{\pi(1+u)} = \frac{-\sin(\pi u)}{\pi(1+u)}$$

We expand both the sine term and the rational term into their Maclaurin series in terms of u:

$$\sin(\pi u) = \pi u - \frac{(\pi u)^3}{6} + O(u^5)$$

$$\frac{1}{1+u} = 1 - u + u^2 - u^3 + O(u^4)$$

Multiplying these expansions together:

$$f(1+u) = -\frac{1}{\pi}\left(\pi u - \frac{\pi^3 u^3}{6} + \dots\right)(1 - u + u^2 - u^3 + \dots)$$

$$f(1+u) = -\left(u - \frac{\pi^2 u^3}{6} + \dots\right)(1 - u + u^2 - u^3 + \dots)$$

$$f(1+u) = -\left(u(1 - u + u^2) - \frac{\pi^2 u^3}{6}\right) + O(u^4) = -u + u^2 - u^3 + \frac{\pi^2 u^3}{6} + O(u^4)$$

The general Taylor series of $f(x)$ at $x = 1$ is written as:

$$f(1+u) = f(1) + f'(1)u + \frac{f''(1)}{2}u^2 + \dots$$

Matching the coefficients of the u^2 term between our expansion and the general Taylor series:

$$\frac{f''(1)}{2} = 1 \implies f''(1) = 2$$

Physics & Engineering

4.1 Power Systems

Nominal-π Transmission Line Model

A symmetrical 3-phase, 22 kV, 50 Hz overhead transmission line is 40 km long and delivers a total load of 20 MW at a lagging power factor of 0.85 to a balanced receiving-end. The line parameters per phase per kilometre are: $R = 0.15\,\Omega/\text{km}$, $X_L = 0.40\,\Omega/\text{km}$, $Y = j3.5 \times 10^{-6}\,\text{S/km}$. Using the Nominal-$\pi$ model, calculate the magnitude of the sending-end line-to-line voltage $V_{S,LL}$ (in kV) required to two significant figures.

Solution: First, determine the total lumped line parameters for the 40 km length: - Series Impedance $Z = (R + jX_L) \times l = (0.15 + j0.40) \times 40 = 6 + j16\,\Omega$. - Total Shunt Admittance $Y_{tot} = Y \times l = j3.5 \times 10^{-6} \times 40 = j1.4 \times 10^{-4}\,\text{S}$. In the Nominal-$\pi$ model, half the admittance is placed at each end: $\frac{Y_{tot}}{2} = j0.7 \times 10^{-4}\,\text{S}$. Working per phase, taking the receiving-end voltage as the reference phasor:

$$V_R = \frac{22,000}{\sqrt{3}} \approx 12,701.7\angle 0^\circ \text{ V}$$

The receiving-end load current I_R is derived from the total 3-phase power $P = 20$ MW:

$$I_R = \frac{P}{\sqrt{3} V_{LL} \cos\phi} = \frac{20 \times 10^6}{\sqrt{3} \times 22,000 \times 0.85} \approx 617.48 \text{ A}$$

The power factor angle is $\phi = \arccos(0.85) \approx -31.79^\circ$ (negative for lagging):

$$I_R = 617.48(\cos(-31.79^\circ) + j\sin(-31.79^\circ)) = 524.86 - j325.29 \text{ A}$$

The current through the receiving-end shunt admittance is:

$$I_{Y,R} = V_R \times \frac{Y_{tot}}{2} = 12701.7 \times (j0.7 \times 10^{-4}) = j0.889 \text{ A}$$

The total series line current is $I_L = I_R + I_{Y,R}$:

$$I_L = 524.86 - j325.29 + j0.889 = 524.86 - j324.4 \text{ A}$$

The per-phase sending-end voltage V_S is found by adding the series voltage drop:

$$V_S = V_R + I_L Z = 12701.7 + (524.86 - j324.4)(6 + j16)$$

$$V_S = 12701.7 + (3149.16 + 5190.4) + j(8397.76 - 1946.4) = 21041.3 + j6451.4 \text{ V}$$

The magnitude of the phase voltage is:

$$|V_S| = \sqrt{21041.3^2 + 6451.4^2} \approx 22,008 \text{ V}$$

Converting back to the line-to-line sending-end voltage:

$$|V_{S,LL}| = \sqrt{3}|V_S| = \sqrt{3} \times 22008 \approx 38,118 \text{ V} = 38.1 \text{ kV}$$

Rounding to two significant figures, we get:

4.2 Quantum Mechanics

Angular Momentum Expectation

Suppose a particle is in the $l = 1$ state and is in the normalized state

$$|\psi\rangle = \frac{4}{\sqrt{18}}|1,1\rangle + \frac{i}{\sqrt{18}}|1,0\rangle + \frac{1}{\sqrt{18}}|1,-1\rangle$$

The expectation value of $L_x^2 + L_y^2$ is $k\hbar^2$. Find the exact value of k.

Solution: The total orbital angular momentum operator squared is given by $L^2 = L_x^2 + L_y^2 + L_z^2$. We can rearrange this to find the operator in question:

$$L_x^2 + L_y^2 = L^2 - L_z^2$$

When acting on an eigenstate $|l,m\rangle$, the eigenvalue equations are $L^2|l,m\rangle = l(l+1)\hbar^2|l,m\rangle$ and $L_z^2|l,m\rangle = m^2\hbar^2|l,m\rangle$. Thus, for any state $|1,m\rangle$ with $l = 1$:

$$(L_x^2 + L_y^2)|1,m\rangle = (1(1+1)\hbar^2 - m^2\hbar^2)|1,m\rangle = (2-m^2)\hbar^2|1,m\rangle$$

We calculate the expectation value using the probabilities $P_m = |c_m|^2$:

$$\langle\psi|(L_x^2 + L_y^2)|\psi\rangle = \sum_{m=-1}^{1} |c_m|^2(2-m^2)\hbar^2$$

Determining the probabilities from the given normalized wavefunction coefficients: - For $m = 1$: $c_1 = \frac{4}{\sqrt{18}} \implies P_1 = \frac{16}{18}$. The eigenvalue is $(2-1^2)\hbar^2 = 1\hbar^2$. - For $m = 0$: $c_0 = \frac{i}{\sqrt{18}} \implies P_0 = \left|\frac{i}{\sqrt{18}}\right|^2 = \frac{1}{18}$. The eigenvalue is $(2-0^2)\hbar^2 = 2\hbar^2$. - For $m = -1$: $c_{-1} = \frac{1}{\sqrt{18}} \implies P_{-1} = \frac{1}{18}$. The eigenvalue is $(2-(-1)^2)\hbar^2 = 1\hbar^2$. Summing these contributions:

$$\langle L_x^2 + L_y^2\rangle = \left(\frac{16}{18}(1) + \frac{1}{18}(2) + \frac{1}{18}(1)\right)\hbar^2 = \frac{16+2+1}{18}\hbar^2 = \frac{19}{18}\hbar^2$$

Thus, the constant k is exactly:

$$\frac{19}{18}$$

4.3 Mechanics

Conservation of Angular Momentum

A horizontal circular platform of radius $R = 0.5$ m and mass $M = 0.45$ kg is free to rotate. Two massless spring toy-guns, each carrying a steel ball of mass $m = 0.05$ kg, are at a distance $r = 0.25$ m from the centre. The guns simultaneously fire the balls horizontally and perpendicular to the diameter in opposite directions. The balls have a horizontal speed of $v = 9$ m s^{-1} with respect to the ground. Find the rotational speed ω (in rad s^{-1}) of the platform.

Solution: We analyze the system using the principle of conservation of angular momentum. Initially, the platform and the balls are at rest, so the total initial angular momentum is $L_{initial} = 0$. When the balls are fired, they travel in opposite directions perpendicular to the diameter. This setup creates a couple, imparting an angular momentum to the balls relative to the center of rotation. Because there are no external torques, the platform must rotate in the opposite direction to conserve angular momentum. The angular momentum of the two balls is:

$$L_{balls} = 2 \times (mvr)$$

$$L_{balls} = 2 \times (0.05 \text{ kg}) \times (9 \text{ m/s}) \times (0.25 \text{ m}) = 0.1 \times 2.25 = 0.225 \text{ kg m}^2\text{s}^{-1}$$

The platform is a uniform circular disk, so its moment of inertia I_p is:

$$I_p = \frac{1}{2}MR^2 = \frac{1}{2}(0.45 \text{ kg})(0.5 \text{ m})^2 = 0.5 \times 0.45 \times 0.25 = 0.05625 \text{ kg m}^2$$

The final angular momentum of the system is the sum of the angular momenta of the balls and the platform, which must equal zero. Letting the platform spin in the opposite direction with angular speed ω:

$$I_p\omega - L_{balls} = 0 \implies I_p\omega = L_{balls}$$

$$0.05625 \times \omega = 0.225$$

Solving for ω:

$$\omega = \frac{0.225}{0.05625} = 4$$

The rotational speed of the platform is 4 rad s^{-1}.